

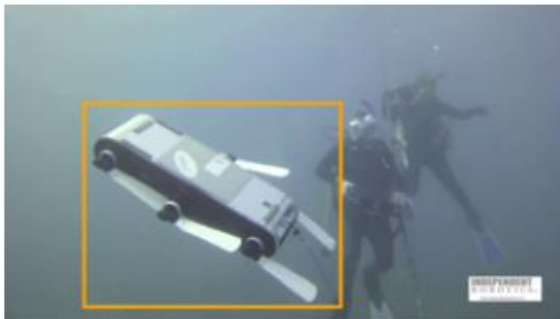
ProMi: An Efficient Prototype-Mixture Baseline for Few-Shot Segmentation with Bounding-Box Annotations

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Paper: <https://arxiv.org/pdf/2505.12547>

Code: <https://github.com/ThalesGroup/promi>



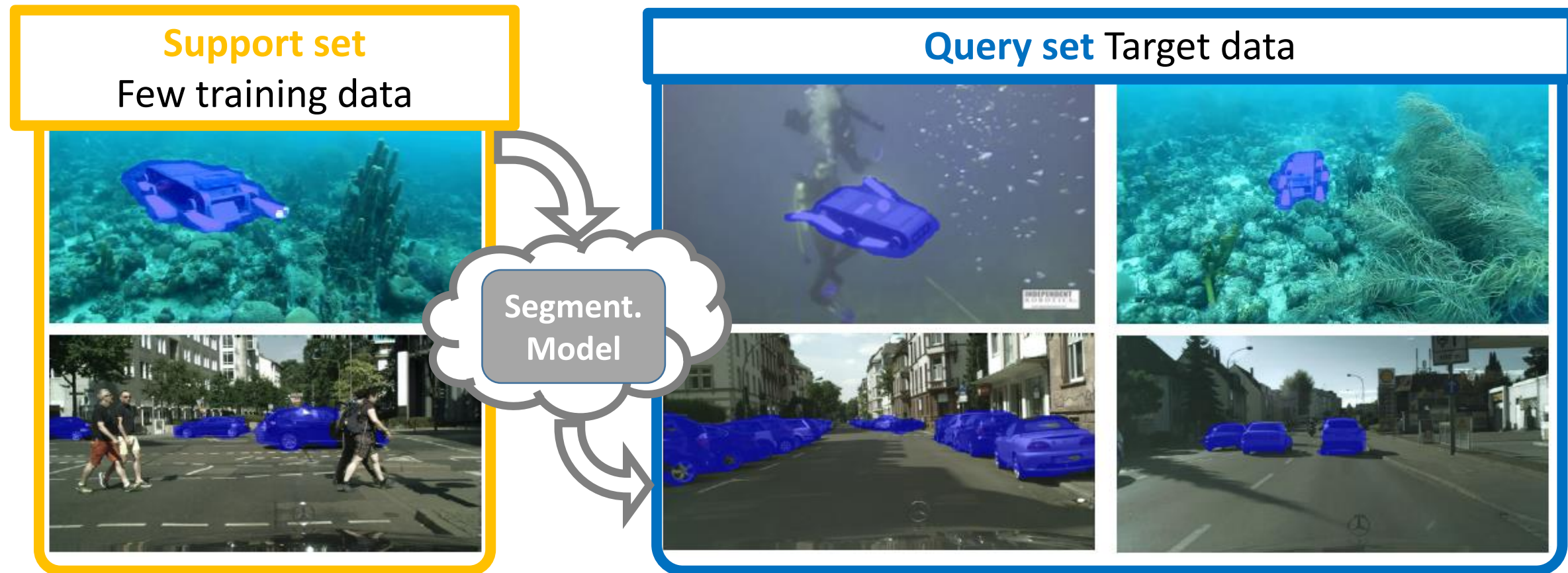
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- **Challenge:** Image Segmentation



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- **Challenge:** **Few-Shot Segmentation**



ProMi: An Efficient Prototype-Mixture Baseline for Few-Shot Segmentation with Bounding-Box Annotations

- **Challenge:** Few-Shot Segmentation

**Manual pixel-level labeling
TAKES TIME**



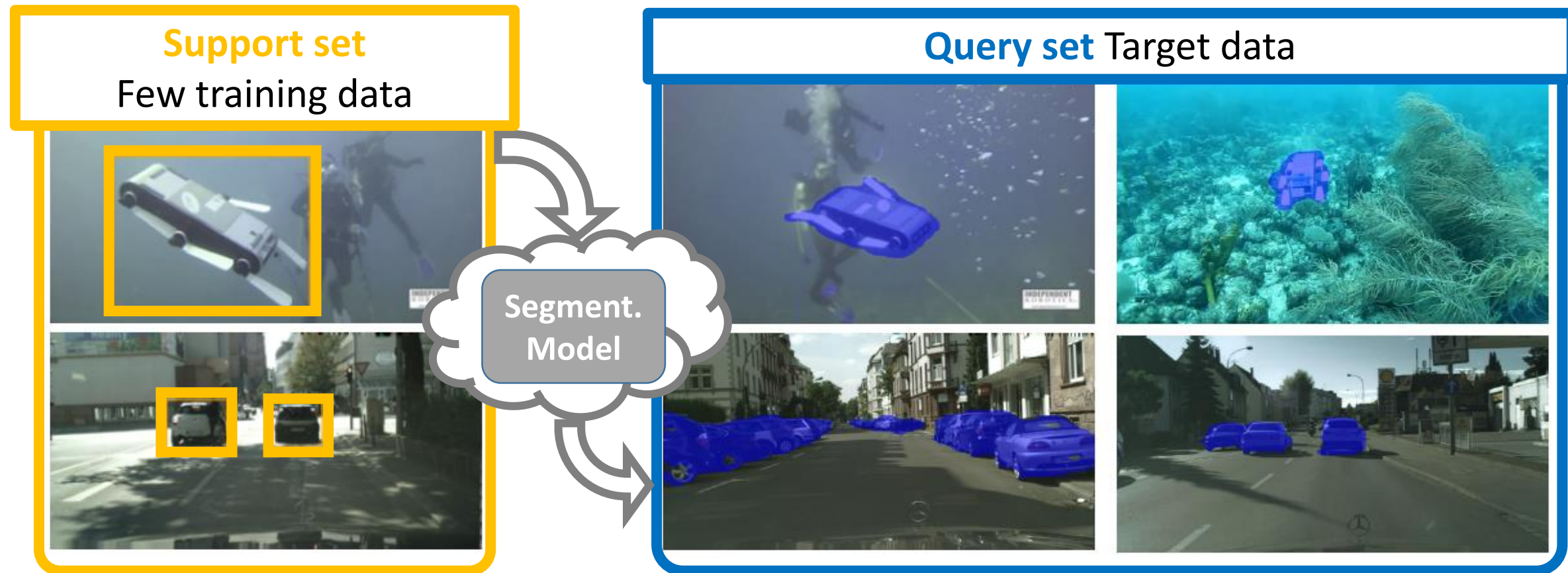
Segment.
Model

Query set Target data



ProMi: An Efficient Prototype-Mixture Baseline for Few-Shot Segmentation with Bounding-Box Annotations

- **Challenge:** **Few-Shot Segmentation with Bounding-Box labels**

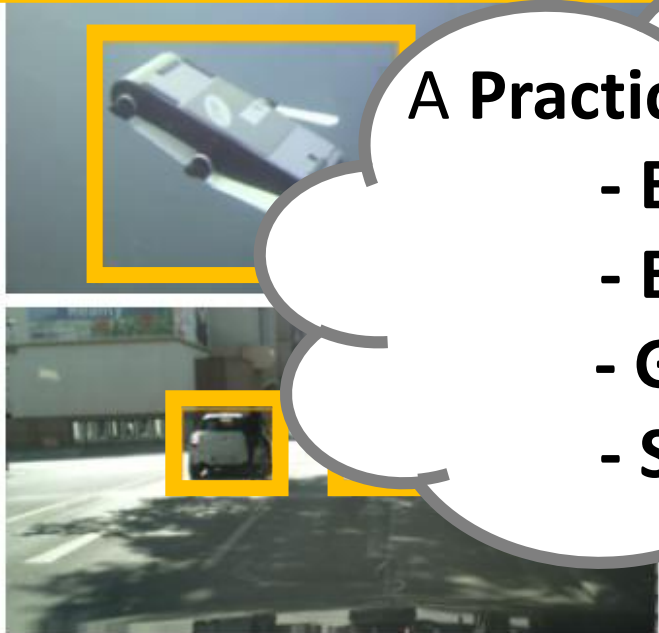


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• Challenge: Few-Shot Segmentation with Bounding-Box labels

Support set

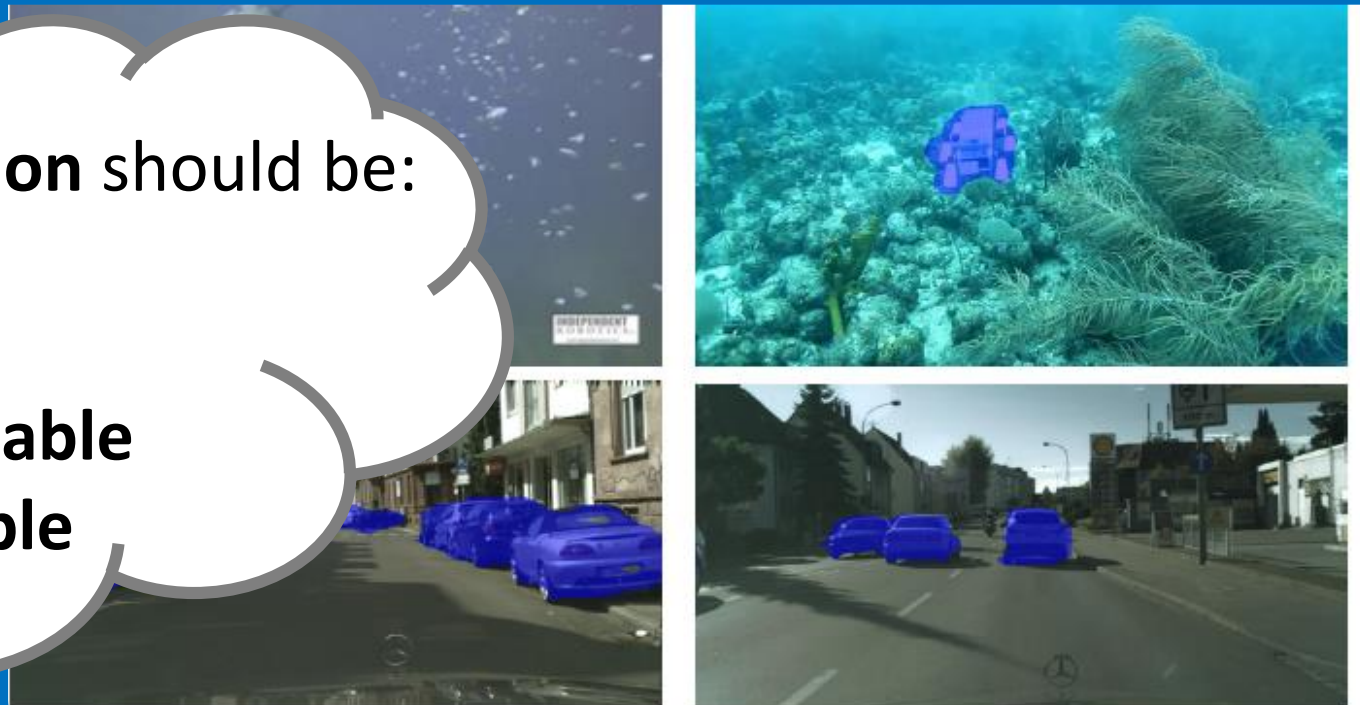
Few training data



A Practical Solution should be:

- Effective
- Efficient
- Generalizable
- Sustainable

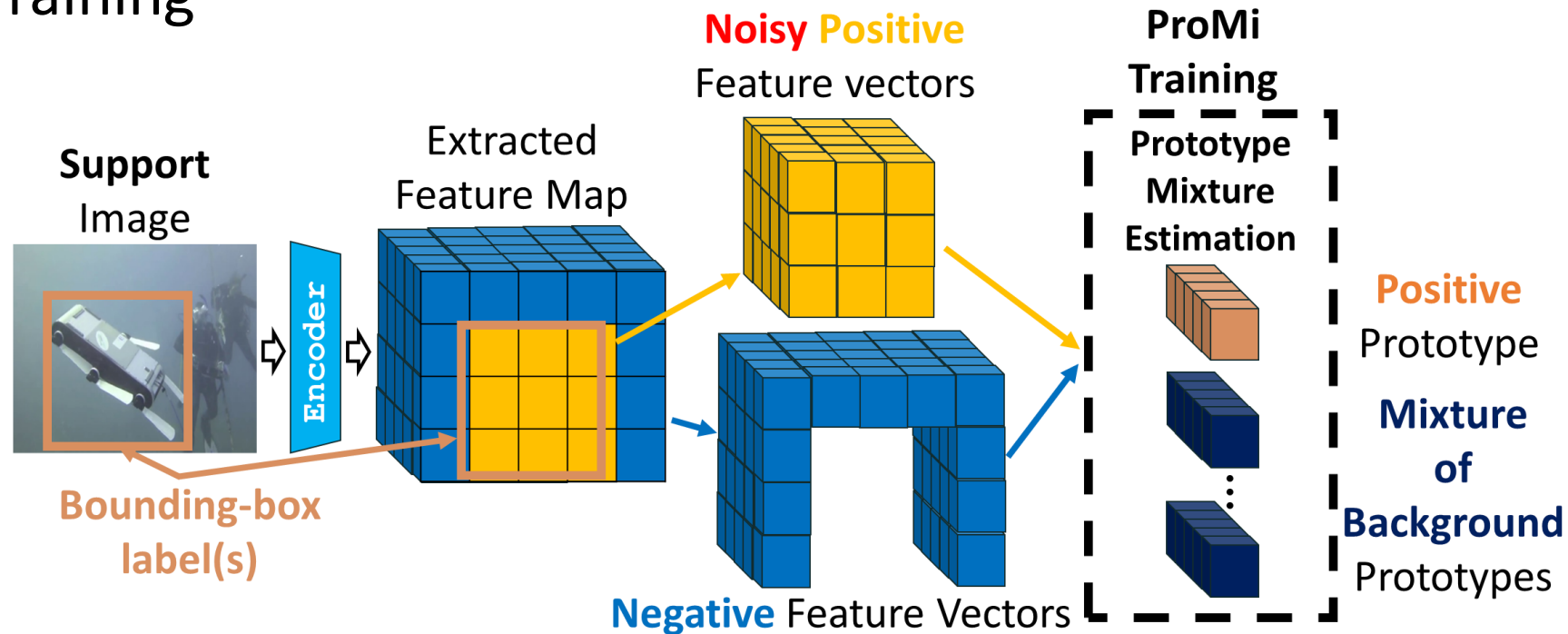
Query set Target data



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- **Proposed Framework => Pre-trained Image Encoder + Latent Space Learning**

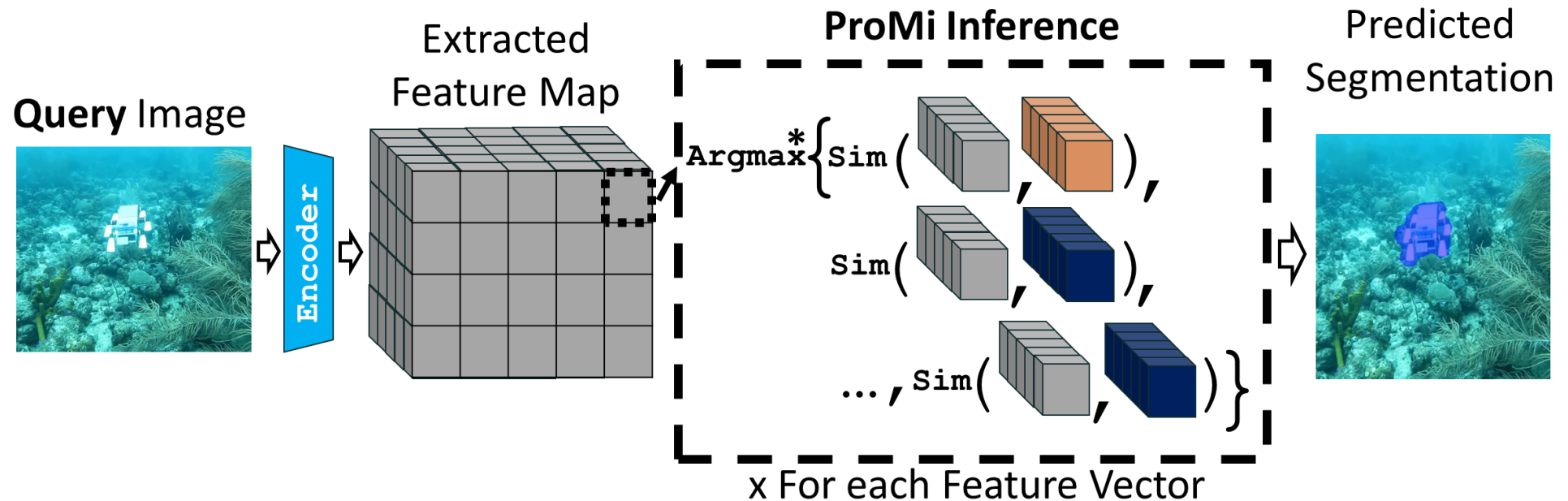
A. Training



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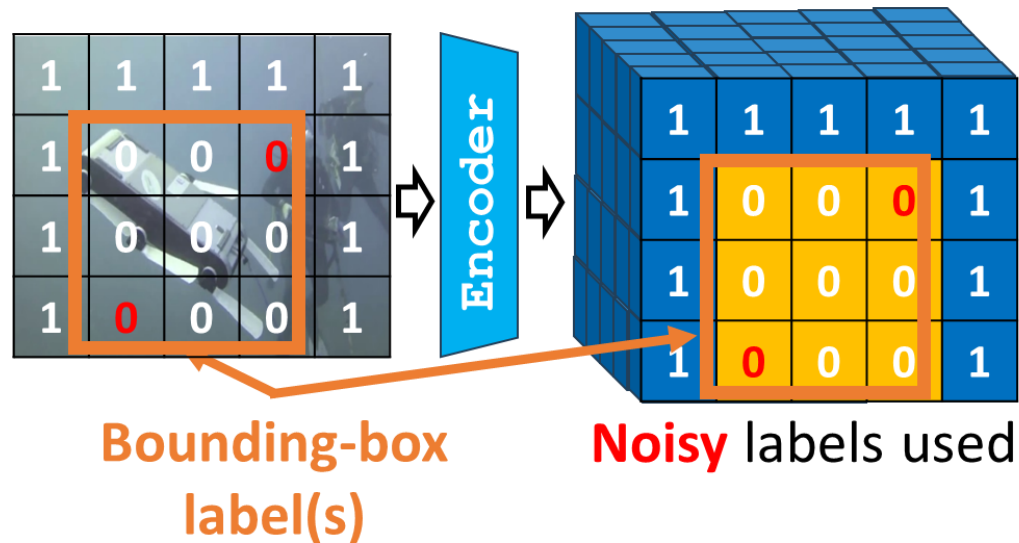
- **Proposed Framework** => **Pre-trained Image Encoder + Latent Space Learning**

B. Inference



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- **Prototype-Mixture Estimation**

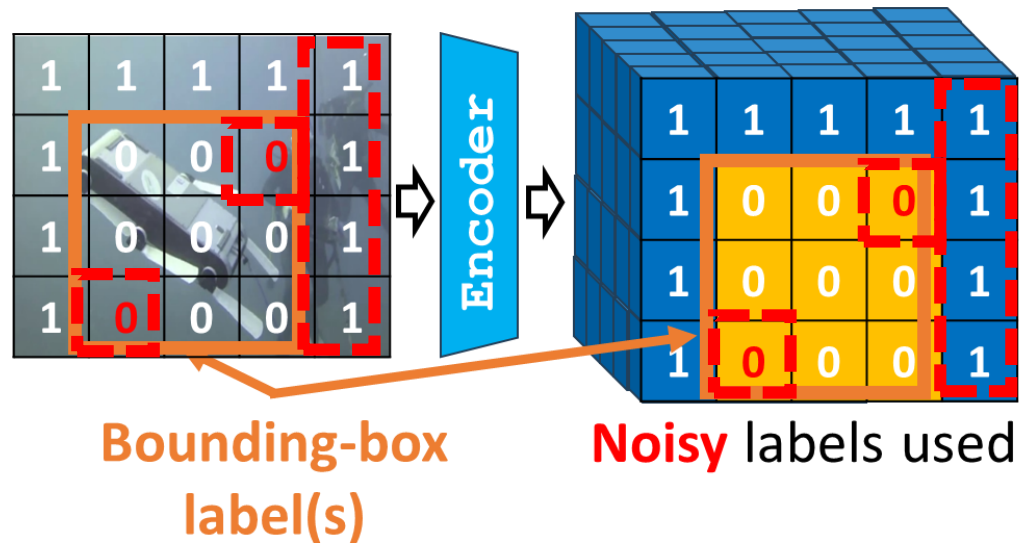


0: FG Class **Robot** | 1: BG Class

ProMi: An Efficient Prototype-Mixture Baseline for Few-Shot Segmentation with Bounding-Box Annotations

- **Prototype-Mixture Estimation**

=> Consider **Noisy Positive Labels** and **Background Mixture distribution**

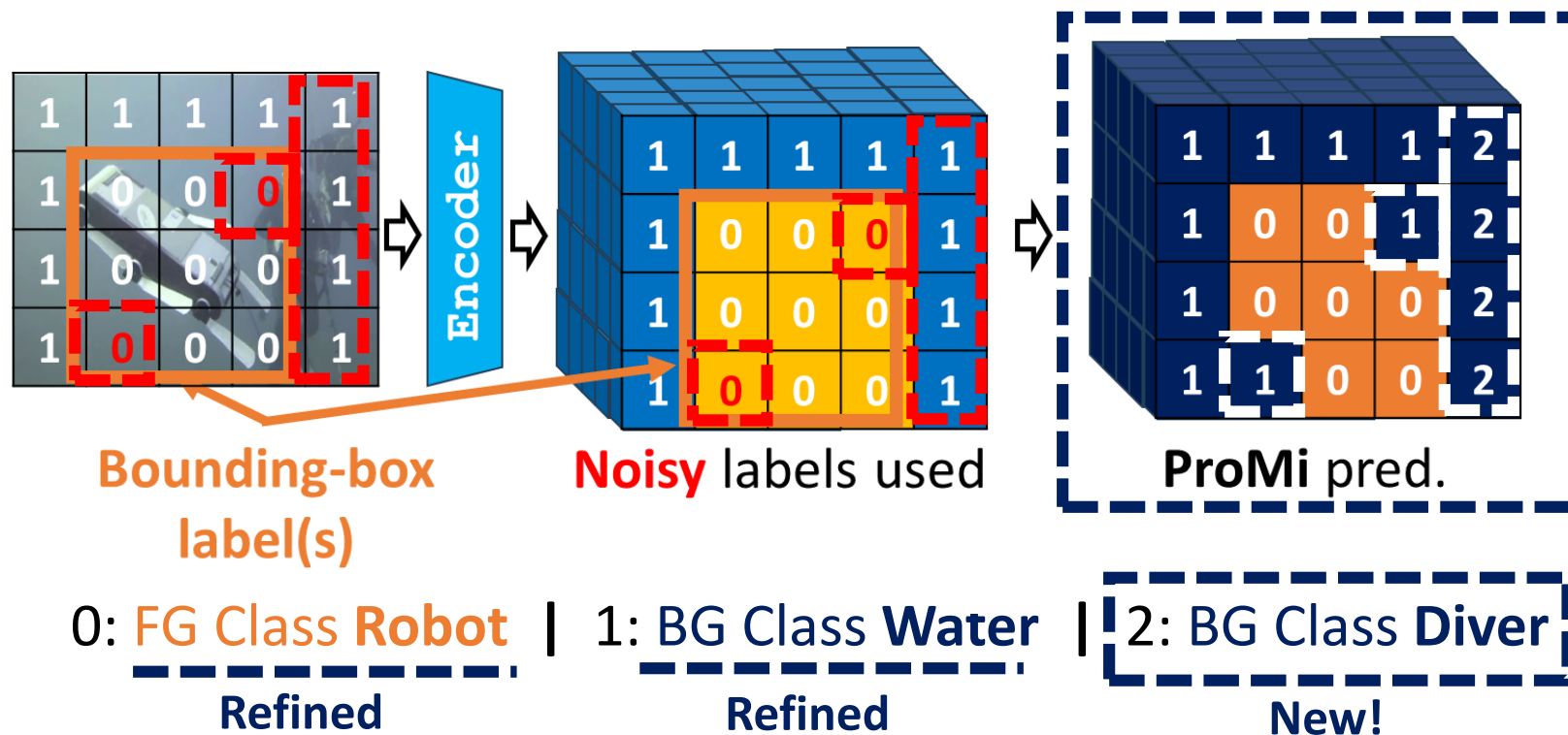


0: FG Class **Robot** | 1: BG Class

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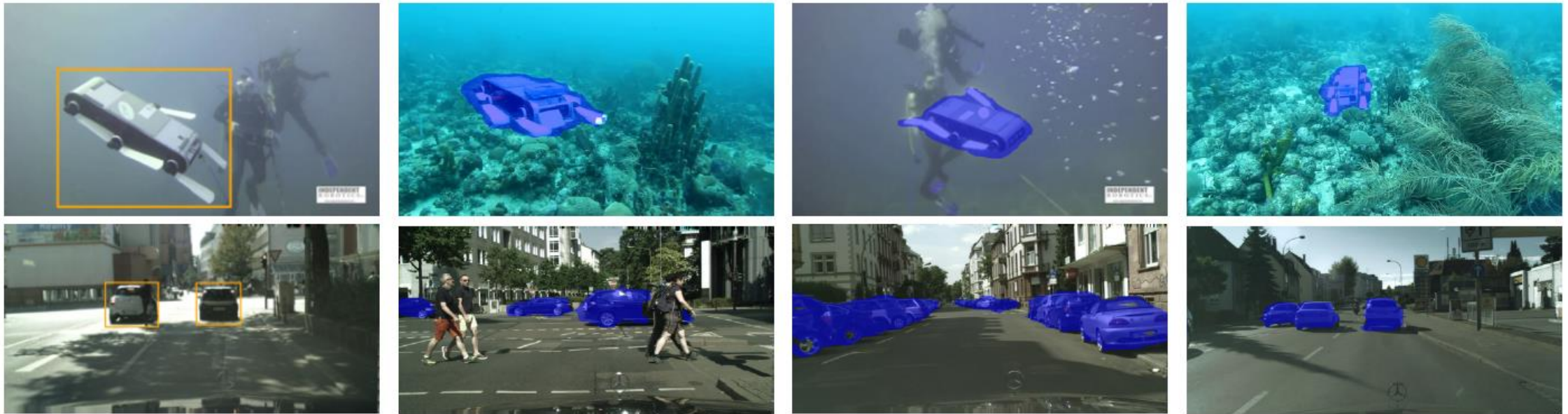
- **Prototype-Mixture Estimation**

=> Consider **Noisy Positive Labels** and **Background Mixture distribution**



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- **Results**



One-Shot Annotation

Predictions

Figure 1. One-Shot Segmentation with Bounding-Box Annotations using ProMi.

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- **Results**



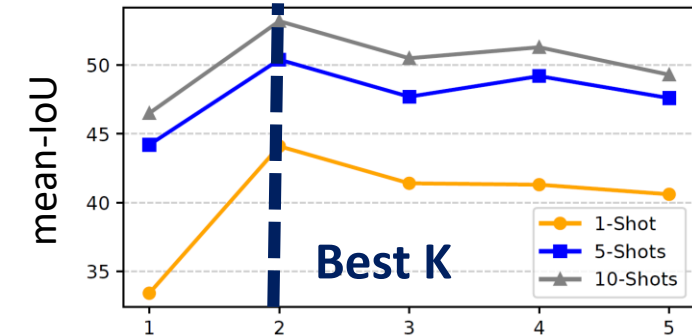
Figure 2. One-Shot result on **high-resolution** UAV data.

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• Quantitative Results

Method	Backbone	PASCAL VOC 2012			MS-COCO		
		1-Shot	5-Shot	10-Shot	1-Shot	5-Shot	10-Shot
BD-CSPN [21]	ResNet-50	43.9	45.9	45.9	23.4	25.5	25.7
SimpleShot [43]	ResNet-50	41.4	44.8	45.5	22.8	25.5	25.7
ProMi (proposed)	ResNet-50	45.4	51.5	55.8	26.1	31.2	33.6
BD-CSPN [21]	DINOv2	39.6	44.3	45.9	18.1	22.4	23.4
SimpleShot [43]	DINOv2	36.7	43.7	45.8	16.5	22.3	23.9
ProMi (proposed)	DINOv2	44.1	50.4	53.2	20.7	28.4	31.2

Table 1. Quantitative comparison (mean-loU).



Number of background (BG) prototypes (K)

Figure 3. Finding the **optimal** number of background (BG) prototypes (K).

BG Mixt.	FG Ref.	1-Shot	5-Shot	10-Shot
×	×	33.4	44.2	46.5
✓	×	40.7	46.0	48.4
✓	✓	44.1	50.4	53.2

Table 2. Ablation of ProMi (mean-loU).

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- **Qualitative Ablation Results: Baseline vs. ProMi**

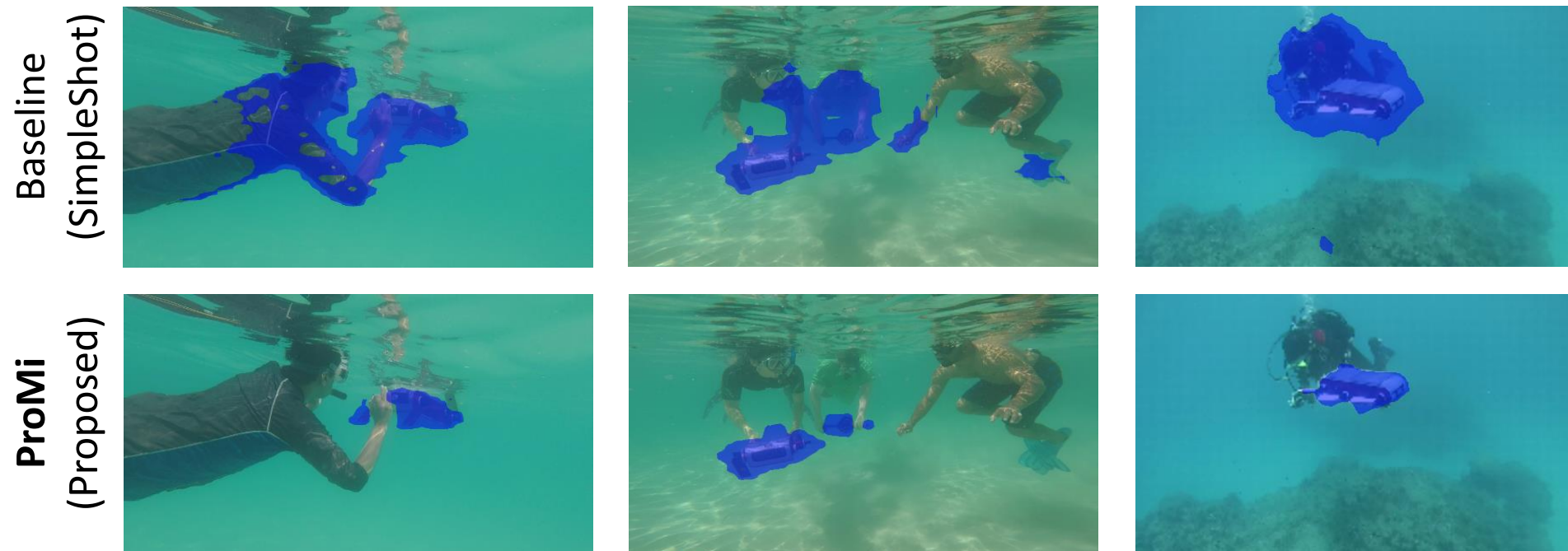


Figure 4. Qualitative comparison between SimpleShot baseline (Top), and ProMi (Bottom) for underwater **robot segmentation**.

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- **Conclusion**

- **Strengths**

- **Effective:** Strong {One, few}-shot results
 - **Efficient training:** Latent space focus
 - **Generalizable:** Applicable across different domains, viewpoints, and tasks
 - **Sustainable:** Backbone-agnostic
 - **Practical Demo Code here:** <https://github.com/ThalesGroup/promi>

- **Limitations**

- Focus on **single foreground class** scenarios
 - **Segmentation Resolution** depends on Encoder Resolution

- **Ongoing & Future Work**

- **Multiclass Extension**
 - **Efficient Post-Processing** for high resolution segmentation (Ongoing promising results)
 - **Extension to Black-Box Challenges** [1] (without feature map access)

[1] Florent et al., *Simplex via sBeta with Applications to Online Adjustment of Black-Box Predictions*. IEEE TPAMI, 2024.